Data Analysis Notes

Data Quality and Profiling

Contents

1 Introduction to Data Analysis and Data Quality

Data analysis plays a **pivotal role in business decision-making and operational processes**. Organizations rely on data to make informed, confident, and strategic decisions. However, the value of these decisions depends heavily on the **quality of the data** being used.

For data to be useful:

- The information must be **relevant to the project objectives**
- The dataset must be **free from errors and inconsistencies**
- The data should be **available in the required format and time**

One of the most challenging aspects of data analysis—whether working with small datasets or spreadsheets containing thousands of rows—is ensuring that the data remains **clean and trustworthy** throughout the analysis process.

2 Data Profiling

2.1 What is Data Profiling?

Data profiling is the process of **examining, inspecting, and summarizing data** to identify:

- Inconsistencies
- Missing values
- Formatting issues
- Duplicate records

The goal of data profiling is to **understand the structure and condition of the data** before performing deeper analysis. This step allows analysts to detect issues early and prevent incorrect conclusions later.

3 Challenges in Maintaining Clean Data

Finding and maintaining clean data is one of the most difficult tasks in data analysis. Common challenges include:

- Human data entry errors
- Inconsistent data formats

- Missing or incomplete information
- Outdated records

To systematically address these challenges, data analysts evaluate datasets using **five key traits of data quality**.

4 Five Key Traits of High-Quality Data

4.1 Accuracy

Accuracy is the **most critical aspect of data quality**. Accurate data correctly represents real-world values without errors.

Common steps to improve accuracy include:

- Removing duplicate records
- Correcting formatting errors (e.g., dates, currency)
- Eliminating blank or invalid rows

Without accurate data, any analysis or prediction derived from it becomes unreliable.

4.2 Completeness

Completeness refers to whether **all required information is available** in the dataset.

Example: If the task is to calculate total sales revenue by region, but the dataset does not include region information, the data is considered incomplete.

Why completeness matters:

- Missing data can invalidate the analysis
- Additional data sources may be required
- Assumptions made due to missing data increase risk

4.3 Reliability

Reliability measures whether data can be **trusted consistently over time**.

Example: If sales agents maintain personal records and fail to update the shared company database regularly, the centralized data becomes unreliable.

Key indicators of unreliable data:

- Inconsistent updates
- Conflicting data sources

- Lack of standardized data entry processes

In such cases, organizations must establish better processes to ensure reliable data collection.

4.4 Relevance

Relevance ensures that the data collected is **necessary and useful** for the specific project.

Example: When analyzing sales revenue per customer, including customer birthdays or personal details adds no analytical value.

Benefits of ensuring relevance:

- Reduces data processing time
- Simplifies analysis
- Avoids privacy and compliance issues

4.5 Timeliness

Timeliness refers to whether data is **current, accessible, and updated at the required frequency**.

Example: If a sales report is used for weekly employee reviews but updated only once per month, the data becomes outdated.

Consequences of poor timeliness:

- Incorrect performance evaluations
- Poor business decisions
- Loss of trust in analytics systems

5 Role of a Data Analyst in Ensuring Data Quality

A data analyst is responsible for:

- Profiling and cleaning data
- Evaluating datasets against quality traits
- Identifying risks caused by poor data
- Ensuring the data is fit for decision-making

By carefully applying the five traits of data quality, analysts can:

- Save time
- Avoid serious analytical errors
- Produce reliable and actionable insights

6 Next Steps

After qualifying and cleaning the data, the next step in the data analysis workflow is to **import the collected data into a spreadsheet or analysis tool**. This prepares the dataset for exploration, visualization, and modeling.

7 Importing Text and CSV Data into Excel

After understanding the importance of data quality, the next step in data analysis is learning how to **import external data into Excel**. Excel primarily works with `.xlsx` and `.xls` files, but it can also import data from other formats such as **text files** and **CSV (Comma-Separated Values) files**.

Kaggle car-sales`.xls`(comma separated) file use for this practice.

7.1 Text and CSV File Formats

Text files may use different characters to separate data fields, such as:

- Commas (CSV files)
- Tabs
- Other delimiters

In a CSV file, each record appears on a new line, and individual values are separated by commas. The first row often contains **column headings**, which should be used as headers when importing into Excel.

7.2 Using the Text Import Wizard

To open a text or CSV file in Excel:

1. Go to **File** → **Open**
2. Select the file or click **Browse**
3. Excel automatically launches the **Text Import Wizard**

Excel detects whether the file is **delimited** (data separated by commas or tabs). Since the first row contains headings, the option **"My data has headers"** must be selected.

7.3 Selecting Delimiters

In the second step of the Text Import Wizard:

- Choose the correct delimiter (e.g., **Comma**)
- Deselect any unnecessary delimiters
- Review the data preview to ensure correct column separation

The preview window helps confirm that the data will appear as expected once imported.

7.4 Setting Column Data Formats

In the final step of the wizard, Excel allows the user to define the **data type** for each column, such as:

- General
- Text
- Date

If no special formatting is required, the default **General** format can be used to complete the import.

7.5 Adjusting Column Widths and Row Heights

After importing, some columns may not display data fully, showing hash symbols (###). This occurs when column widths are too narrow.

To fix this:

- Select all columns and double-click any column divider to auto-fit
- Manually drag column or row borders to resize
- Double-click row dividers to auto-size row height

7.6 Adding and Removing Columns and Rows

During data preparation, unnecessary fields may need to be removed and new ones added.

Removing columns or rows:

- Select the column or row
- Right-click and choose **Delete**

- Or use **Home** → **Cells** → **Delete**

Adding columns or rows:

- Select the column or row next to the desired location
- Right-click and choose **Insert**
- Rename the new column header as needed (e.g., *Year*)

7.7 Saving the File as an Excel Workbook

After editing, the file can be saved as an Excel workbook by:

- Choosing **File** → **Save As**
- Selecting **Excel Workbook (*.xlsx)** as the file type

Excel may also display a notification allowing quick conversion to the Excel format.

7.8 Summary

In this section, we learned how to:

- Import text and CSV files using the Text Import Wizard
- Select delimiters and column formats
- Adjust column widths and row heights
- Add and remove columns and rows

These steps ensure that imported data is properly structured and ready for further analysis. The next topic will focus on **data privacy**, including sensitive and personally identifiable information.

8 Data Privacy and Regulatory Compliance

As data analysts work with increasing amounts of customer and organizational data, **data privacy** becomes a critical responsibility. Data privacy refers to the rules and practices that govern how collected data can be accessed, used, stored, and shared. Understanding these regulations helps organizations **avoid legal penalties**, **protect individuals**, and **maintain customer trust**.

8.1 Core Fundamentals of Data Privacy

Data privacy is built on three essential fundamentals:

1. **Confidentiality**
2. **Collection and Use**
3. **Compliance**

A clear understanding of these fundamentals allows analysts to handle data responsibly and ethically.

8.2 Confidentiality

Confidentiality recognizes that **personal data belongs to the individual**, not the organization collecting it. Data analysts may work with various types of information, such as:

- Sales and revenue forecasts
- Employee records
- Patient or healthcare data

When accessing such data, analysts must be able to identify and protect different categories of personal information.

8.3 Types of Personal Data

Personal data can be classified into three main categories, each with increasing levels of sensitivity:

Data Type	**Description and Examples**
Personal Information (PI)	Any information that can be linked to an individual, such as email addresses, phone numbers, usernames, or images.
Personally Identifiable Information (PII)	Information that can directly identify a specific individual, including passport numbers, national identification numbers, or driver's license numbers.
Sensitive Personal Information (SPI)	Highly private data that may not directly identify a person but could cause serious harm if exposed, such as race, sexual orientation, biometric data, or genetic information.

Understanding these categories allows analysts to **apply appropriate protection levels** and decide what data can be safely removed or anonymized.

8.4 Anonymization and Trust

Anonymization involves removing all information that can link data back to an individual. While not always practical, anonymizing unnecessary fields:

- Reduces privacy risks
- Increases data usability
- Builds consumer confidence

This practice supports the ethical and secure flow of information.

8.5 Geographic Importance of Data Privacy

Data privacy regulations depend heavily on:

- The location of the organization collecting the data
- The location of the individual whose data is collected

Knowing where data originates determines which laws apply and how the data must be handled.

8.6 Major Data Privacy Regulations

Different regions and industries follow different regulations:

- **GDPR (European Union):** Applies to individuals within the EU, regardless of where the data processor is located.
- **LGPD (Brazil):** Applies to individuals in Brazil and focuses on protecting personal data rights.
- **United States:** No single federal law; individual states create their own regulations.
- **CCPA (California):** Protects consumer data for residents of California.

8.7 Industry-Specific Regulations

Some industries are governed by additional privacy standards:

- **Healthcare:** HIPAA regulates protected health information.
- **Retail and Finance:** PCI standards govern credit card data.

Failure to comply with these standards can result in severe financial penalties.

8.8 Data Breaches and Organizational Responsibility

Organizations are responsible for safeguarding customer data at all times. Consider the following scenario:

> A data analyst downloads sensitive data and takes their work laptop home. The laptop is stolen from their car, resulting in unauthorized access to customer information.

Even though the loss was accidental, this situation constitutes a **data privacy breach**. Such incidents can:

- Lead to large fines and legal penalties
- Damage brand reputation
- Reduce customer trust and revenue

8.9 When Data Privacy Laws Do Not Apply

Data privacy regulations generally do not apply when data is:

- Fully anonymous
- Impossible to trace back to any individual

Although complete anonymization is not always feasible, collecting data with privacy considerations from the beginning can reduce regulatory limitations.

8.10 Summary

In this section, we learned:

- The importance of data privacy in analytics
- The three fundamentals: confidentiality, collection and use, and compliance
- Key types of personal data and their risks
- Global and industry-specific data privacy regulations

A strong understanding of data privacy enables analysts to work responsibly and protect both individuals and organizations. The next lesson will focus on **methods for cleaning data in spreadsheets**.

9 Why Data Quality and Data Privacy Matter in Data Analysis

In this lesson, experienced data professionals share real-world insights on why **data quality** and **data privacy** are fundamental to successful data analysis. Their perspectives highlight how poor data practices can undermine trust, mislead decision-making, and reduce the impact of analytical work.

9.1 Importance of Data Quality in Data Analysis

Data quality forms the **backbone of any successful data analysis project**. Analysts are often questioned when their findings do not align with business expectations:

- Where did the data come from?
- How was it transformed?
- Can the source be trusted?

Since stakeholders believe they understand their business well, any contradiction requires the analyst to rely on **clean, accurate, and trusted data**. Without this foundation, discussions can turn into debates, and the key message of the analysis may be lost.

Garbage In, Garbage Out

A well-known principle in computer science is **"Garbage In, Garbage Out"**. This means:

> If poor-quality data is used as input, the resulting analysis will also be poor.

Analyzing inaccurate or inconsistent data wastes time and can lead to incorrect conclusions. Therefore, analysts must always **sense-check the data** and ensure they are confident in its accuracy before proceeding.

9.2 Impact of Poor Data Quality on Business Decisions

Data is used to guide critical business decisions such as:

- Product launches
- Profitability analysis
- Inventory management

Example: If an analyst selects the wrong Stock Keeping Unit (SKU) during an inventory analysis, they may conclude that a profitable product is underperforming. This mistake could lead to poor strategic decisions, financial loss, and reduced trust in analytics.

Once flawed data is used as a foundation, correcting errors later means losing:

- Time
- Effort
- Credibility

9.3 Importance of Data Privacy in Data Analysis

Data privacy is equally critical, particularly in industries such as:

- Healthcare
- Pharmaceuticals
- Finance and taxation

However, data privacy is not limited to regulated industries—it applies across all sectors. Analysts must ensure that users access only the data **appropriate to their role and permissions**.

9.4 Access Control and Role-Based Permissions

Data privacy can be enforced through:

- Geographic data restrictions (e.g., country-level access)
- Functional access controls (e.g., department-specific views)
- Role-based permissions within analytics tools

Some analytics platforms allow data access rules to be built directly into the data model, controlling who can see specific reports or fields.

9.5 Protecting Personally Identifiable Information (PII)

In fields such as taxation and finance, protecting **Personally Identifiable Information (PII)** is essential. Sensitive data such as:

- Social security numbers
- Names

- Dates of birth

must never be shared carelessly.

Best practices include:

- Avoiding email transmission of sensitive data
- Using encryption when sharing files
- Masking or hiding sensitive fields
- Sharing authentication details through separate communication channels

These practices ensure that data remains secure even when shared with clients or stakeholders.

9.6 Key Takeaways

From these professional insights, we learn that:

- High-quality data is essential for trustworthy analysis
- Poor data leads to misleading conclusions and wasted effort
- Data privacy must be enforced across all industries
- Protecting sensitive information is a continuous responsibility

Ensuring strong data quality and privacy practices not only protects organizations from risk but also builds long-term trust in data-driven decision-making.

10 Cleaning Inaccurate, Empty, and Duplicate Data in Excel

After understanding data quality and data privacy, the next essential step in data analysis is **data cleaning**. Errors and inconsistencies frequently occur during data collection and import—whether manual or automated—and must be corrected before meaningful analysis can take place.

10.1 Common Data Errors and Their Impact

Typical data issues include:

- Spelling mistakes
- Extra whitespace or incorrect letter case

- Empty rows or missing values
- Inaccurate or duplicated records

These problems can cause:

- Formulas to fail
- Sorting and filtering errors
- Misleading charts and reports

To prevent these issues, analysts must follow a structured **data-cleaning routine**.

10.2 Spell Checking Data

Spell checking is one of the simplest yet most effective cleaning steps. Excel's spell-checking feature works similarly to Microsoft Word.

Workflow:

1. Select the column to check (e.g., Product Line or Country)
2. Go to **Review** → **Spelling**
3. Review suggestions and choose:
   - **Change** to correct the spelling
   - **Ignore** if the value is correct

This step helps fix typographical errors such as misspelled country names or incorrect category labels (e.g., *Small*, *Medium*).

10.3 Identifying and Removing Empty Rows

Empty rows can disrupt data navigation and analysis. One common sign of empty rows is when the shortcut `CTRL + Down Arrow` stops prematurely instead of jumping to the last row of data.

Why Empty Rows Are a Problem

- Break continuous datasets into fragments
- Cause formula and filter errors
- Reduce analysis efficiency

Efficient Method Using Filters

For large datasets, filtering is the fastest and safest approach:

1. Select the entire dataset (`CTRL + SHIFT + END`)
2. Enable filters: **Data → Filter**
3. Open a column filter (e.g., Customer Name)
4. Uncheck **Select All**, then select **Blanks**

Only empty rows will now be visible.
Next steps:

- Select the highlighted rows
- Delete them
- Clear and turn off the filter

After removal, navigation shortcuts function correctly again.

10.4 Detecting and Removing Duplicate Data

Duplicate rows often appear due to human entry errors or import issues. Excel provides two approaches to handle duplicates.

Method 1: Review Before Deleting (Recommended)

This method allows visual verification before deletion.
Steps:

1. Select a column unlikely to contain duplicates (e.g., *Sales*)
2. Go to **Home → Conditional Formatting**
3. Choose **Highlight Cells Rules → Duplicate Values**

Highlighted entries can then be inspected to confirm they are true duplicates before deleting the extra rows.

Best Practice: Avoid columns like *Price Each*, where duplicate values are expected.

Method 2: Remove Duplicates Tool (Faster but Risky)

This method is quicker but removes duplicates without preview.

1. Select the entire dataset
2. Go to **Data** → **Remove Duplicates**
3. Deselect all columns, then select only the key column (e.g., Sales)
4. Confirm removal

Use this approach only when confident in the selection criteria.

10.5 Correcting Errors with Find & Replace

The **Find & Replace** feature helps correct repeated errors efficiently.

Example Workflow:

1. Open **Home** → **Find & Select**
2. Enter the incorrect value in **Find what**
3. Click **Find All** to review occurrences
4. Enter the corrected value in **Replace with**
5. Select **Replace All**

This is especially useful for fixing customer names or standardized labels across large datasets.

10.6 Summary

In this section, we learned how to:

- Fix inaccurate data using spell check
- Identify and remove empty rows efficiently
- Detect and delete duplicate records safely
- Correct repeated errors using Find & Replace

These cleaning techniques improve data accuracy, usability, and reliability. In the next lesson, we will explore **text case changes, date formatting fixes, and trimming whitespace**.

11 Standardizing Text, Fixing Date Formats, and Trimming Whitespace

After removing inaccurate, empty, and duplicate data, the next stage of data cleaning focuses on **standardization**. In this lesson, we learn how to:

- Change the case of text
- Fix inconsistent date formats
- Remove unwanted whitespace

These steps ensure that data is consistent, readable, and suitable for analysis and visualization.

11.1 Understanding Text Case Issues

When data is collected from multiple sources, text often appears in mixed case:

- UPPERCASE (e.g., `COUNTRY`)
- lowercase (e.g., `product`)
- Proper case (e.g., `Customer Name`)

While some inconsistencies are intentional, many are not. Excel does not have a *Change Case* button like Word, so we use text functions instead.

11.2 Changing Text Case Using Functions

Excel provides three key functions for changing text case:

Function	**Purpose**
`UPPER()`	Converts text to uppercase
`LOWER()`	Converts text to lowercase
`PROPER()`	Converts text to proper (title) case

These functions require the use of a **helper row or helper column**.

11.3 Using the PROPER Function (Headers Example)

If headers are in uppercase and need to be converted to proper case:

1. Insert a new row below the header (helper row)
2. Enter the formula: `=PROPER(A1)`

3. Press **Enter**

To copy across many columns efficiently:

- Select columns using `SHIFT + RIGHT ARROW`
- Press `F2`
- Press `CTRL + ENTER`

Important: Do *not* delete the original header row directly. Instead:

- Copy the helper row
- Paste using **Paste Values**
- Delete the helper row

11.4 Using the UPPER and LOWER Functions (Data Example)

UPPER Function

1. Insert a helper column next to the target column
2. Enter: `=UPPER(T2)`
3. Double-click the Fill Handle to copy down

LOWER Function

1. Insert a helper column
2. Enter: `=LOWER(K2)`
3. Use the Fill Handle to copy down

In both cases:

- Copy the helper column
- Paste **Values** into the original column
- Delete the helper column

11.5 Fixing Date Formatting Issues

Dates may appear inconsistent due to different regional formats.

Example Issue:

- UK format: `DD/MM/YYYY`
- US format: `MM/DD/YYYY`

Changing Locale-Based Date Formats

1. Open the **Number Format** dialog
2. Change **Locale** (e.g., UK to US)
3. Select a preferred date format

Using Custom Date Formats

- Open Number Format → Custom
- Modify an existing format (e.g., `DD-MMM-YYYY`)
- Apply using Format Painter or column selection

11.6 Removing Unwanted Whitespace

Whitespace issues include:

- Leading spaces
- Trailing spaces
- Multiple spaces between words

Using Find & Replace

To remove double spaces:

1. Select the dataset
2. Go to **Home** → **Find & Select** → **Replace**
3. Find: `(double space)`
4. Replace with: `(single space)`

Use **Replace All** only when confident.

Using the TRIM Function

To remove leading and trailing spaces:

1. Insert a helper column
2. Enter: `=TRIM(M2)`
3. Copy down using Fill Handle

Then:

- Copy helper column
- Paste **Values** into original column
- Delete helper column

11.7 Summary

In this section, we learned how to:

- Standardize text using UPPER, LOWER, and PROPER
- Correct inconsistent date formats
- Remove unwanted whitespace using Find & Replace and TRIM

These techniques significantly improve data consistency and reliability. In the next lesson, we will explore **Flash Fill and Text to Columns** to further automate data cleaning.

12 Using Flash Fill and Text to Columns for Data Cleaning

After standardizing text case, fixing date formats, and trimming whitespace, the next powerful tools for data cleaning in Excel are **Flash Fill** and **Text to Columns**. These features help reshape text data quickly and efficiently.

12.1 Flash Fill: Pattern-Based Data Transformation

Flash Fill automatically fills data by **detecting patterns** from your manual input. It is especially useful when working with names, dates, or structured text.

Common Use Cases:

- Combining first name and last name

- Splitting names into components
- Changing name formats (e.g., First Last → Last, First)

Flash Fill works best when:

- Data follows a consistent pattern
- You can provide at least one example

12.2 Example 1: Combining First Name and Last Name

Scenario: You have two columns:

- Column A: First Name
- Column B: Last Name

You want a single column called `ContactName`.

Steps:

1. Insert a new helper column and name it `ContactName`
2. In the first row, manually type the full name (e.g., `John Smith`)
3. Press **Enter**
4. Start typing the next name
5. When Flash Fill shows a preview, press **Enter**

Excel automatically fills the remaining names—even for complex surnames like `Da Cunha`.

Result: Two columns are combined into one. The original columns can be deleted if no longer needed.

12.3 Example 2: Changing Naming Convention

Scenario: You have a column with names in the format:

```
John Smith
```

You want:

```
Smith, John
```

Steps:

1. Insert a new column

2. In the first data cell, type: `Smith, John`
3. Press **Enter**
4. Begin typing the second name
5. Accept the Flash Fill preview by pressing **Enter**

Flash Fill detects the pattern and reformats all names instantly.

12.4 Limitations of Flash Fill

Flash Fill **cannot reliably split a single column into multiple columns**. For that task, Excel provides a dedicated feature: **Text to Columns**.

12.5 Text to Columns: Splitting Text into Multiple Columns

The **Text to Columns** feature breaks multi-part text into separate columns using delimiters such as:

- Space
- Comma
- Tab
- Custom characters

This feature is ideal for:

- Full names
- Addresses
- Codes with separators

12.6 Example: Splitting Full Name into First and Last Name

Scenario: Column A contains full names (e.g., `John Smith`).

Steps:

1. Select the data range (e.g., A2:A23)
2. Go to **Data** → **Text to Columns**
3. Choose **Delimited**
4. Select **Space** as the delimiter

5. Set the destination cell (e.g., B2)
6. Finish the wizard

Result:

- Column B: First Name
- Column C: Last Name

The original column can be deleted if no longer required.

12.7 Using Functions Instead of Text to Columns

In some cases, such as **Excel for the web**, the Text to Columns feature is unavailable. Functions provide a flexible alternative, especially for complex names.

Example Functions:

- `LEFT()` – extracts characters from the left
- `RIGHT()` – extracts characters from the right

Example:

- First Name: `=LEFT(A2,5)`
- Last Name: `=RIGHT(A2,7)`

These formulas can be copied down using the Fill Handle.

Advantage:

- Handles mixed name structures
- Works in Excel desktop and web versions

12.8 Summary

In this section, we learned how to:

- Use **Flash Fill** to combine and reformat text based on patterns
- Use **Text to Columns** to split multi-part text
- Apply text functions as a flexible alternative

These tools greatly reduce manual effort and are essential for professional data cleaning workflows.

13 Shaping and Analyzing Data for Meaningful Insights

After collecting and cleaning data, the next step is deciding **how to analyze it**. Effective analysis requires **filtering, sorting, calculations, and reshaping data** to transform raw data into meaningful information.

13.1 Thinking Before You Analyze

Before making any changes to the dataset, it is important to **visualize the final output**. Key questions to ask include:

- How large is the dataset?
- What filtering is required to find relevant records?
- How should the data be sorted?
- What calculations are needed?

Answering these questions helps determine the best approach to shaping the data.

13.2 Sorting Data

Sorting organizes data based on specific conditions such as:

- Alphabetical order (A–Z or Z–A)
- Numerical order (smallest to largest or vice versa)

Example:

- Sorting by `Order Number` helps quickly identify duplicate orders.

Sorting is often the **first step** in reviewing and validating data quality.

13.3 Filtering Data

Filtering displays **only the rows that meet specific criteria**, hiding all others.
Example:

- Filter the column `MONTH_ID` to show only records where the value equals `11`
- This isolates all data related to the month of November

Filtering allows analysts to focus on **relevant subsets of data** without altering the dataset.

13.4 Performing Calculations Using Functions

Spreadsheets provide built-in **functions** that simplify calculations and reduce errors. Functions are grouped into categories such as:

- Mathematical
- Statistical
- Logical
- Financial
- Date and Time

Example: Average Revenue

- Manual method (error-prone):

```
=B1+B2+B3+...+B160 / 160
```

- Using a function (efficient and accurate):

```
=AVERAGE(B1:B160)
```

Functions make calculations **faster, clearer, and more reliable**.

13.5 Why Convert Data into a Table?

Converting data into an Excel **Table** enhances analysis and usability.

Key Advantages of Tables:

- Automatic calculations that update when filtering
- Column headers remain visible
- Banded rows improve readability
- Tables expand automatically when new rows are added

Example:

- Selecting the `MSRP` column and choosing `Sum` instantly calculates the total
- Filtering the `Country` column to `Japan` recalculates the sum only for Japan

Tables allow calculations to dynamically respond to filters.

13.6 Using Pivot Tables and Pivot Charts

When data becomes too complex for simple tables, **Pivot Tables** offer a powerful way to summarize and analyze large datasets.

Example Use Case:

- Identify which company ordered products in October

Steps:

1. Create a Pivot Table from the original dataset
2. Add a month filter
3. Select `October` as the filter value

The pivot table updates instantly to show only October data.

Pivot Charts visually represent the pivot table data and update automatically when filters change.

13.7 Benefits of Pivot Tables and Charts

- Summarize large datasets quickly
- Manipulate data without complex formulas
- Display clear and engaging visualizations
- Improve audience understanding of trends and patterns

13.8 Summary

In this section, we learned how to:

- Sort and filter data to focus on relevant information
- Use functions to perform accurate calculations
- Convert data into tables for efficient analysis
- Apply pivot tables and pivot charts for advanced data exploration

These tools are essential for transforming raw data into meaningful insights. In the next section, we will explore filtering and sorting in greater depth.

14 Filtering and Sorting Data in Excel

In this section, we learn how to use **Filter** and **Sort** tools in Excel to control

- which data is displayed
- how the data is arranged

These tools help improve data visibility, simplify analysis, and allow analysts to focus only on relevant information.

14.1 Understanding Data Filtering

Filtering allows you to **temporarily hide rows that do not meet specified criteria**, without deleting any data.

Why filtering is important:

- Narrow down large datasets
- Focus on specific values or ranges
- Quickly locate relevant records

14.2 Turning Filters On

To enable filtering:

1. Go to the **Data** tab
2. Click **Filter**

A small filter icon will appear next to each column header.

Notes:

- If you select specific columns before clicking Filter, only those columns will be filterable
- If your data is formatted as a **Table**, filters are added automatically

14.3 Filtering by Column Values

Each column can be filtered independently.

Examples:

- **Orderdate**: Filter by year or month
- **Productline**: Filter by product category

- **Customername**: Filter by specific customer

Example Scenario:

- Filter `Orderdate` to show only orders from **2004**
- Result: Only 50 out of 114 records are displayed (shown in the status bar)

14.4 Clearing Filters

There are two ways to remove filters:

- Clear a single filter using **Clear Filter From ...**
- Clear all filters using the **Clear** button in the Sort & Filter group

14.5 Applying Multiple Filters

You can apply filters on multiple columns at the same time.

Example:

- Year: 2004
- Productline: Classic Cars
- Customer: Mini Gifts Distributors Ltd.

This displays only rows that meet **all filter conditions simultaneously**.

14.6 Using Custom Filters

Custom filters allow you to define **numeric or text-based conditions**.

Number Filters Example (Sales column):

- Show sales **greater than $2000**
- Show sales **less than $2000**

Observation:

- Filtered-out rows are hidden, not deleted
- Blue row numbers indicate hidden rows

For text columns, Excel provides **Text Filters** instead of Number Filters.

14.7 Introduction to Sorting Data

Sorting arranges data in a logical order, making it easier to interpret and analyze.

Common sorting types:

- Text: Alphabetical (A–Z / Z–A)
- Numbers: Smallest to Largest / Largest to Smallest
- Dates: Oldest to Newest / Newest to Oldest

14.8 Basic Sorting Examples

- Sort **Customername** alphabetically
- Sort **Sales** by value
- Sort **Orderdate** chronologically

To sort:

1. Select a cell in the target column
2. Choose the appropriate sort option

14.9 Sorting by Multiple Columns

Excel allows sorting using more than one column (multi-level sorting).

Example:

1. First sort by `Orderdate` (Oldest to Newest)
2. Then sort by `Sales` (Largest to Smallest)

Important:

- Ensure `My data has headers` is checked

This results in:

- Orders grouped by date
- Highest sales shown first within each date

14.10 Key Takeaways

- Filtering controls **what data is visible**
- Sorting controls **how data is ordered**
- Filters do not delete data; they only hide it
- Multi-level sorting helps reveal deeper patterns

Filtering and sorting are foundational skills for effective data analysis and reporting.

15 Why Filtering and Sorting Matter in Data Analysis

In this section, data professionals explain **why filtering and sorting are essential tools** for effective data analysis and visualization.

Filtering and sorting allow analysts to create **one clean, unified view of data** while still giving users the flexibility to explore the data in their own way.

15.1 Purpose of Filtering and Sorting

Filtering and sorting help analysts:

- Reduce information overload
- Focus on relevant subsets of data
- Answer specific business questions quickly
- Support interactive data exploration

Instead of showing all data at once, these tools help reveal only what matters.

15.2 Understanding Sorting

Sorting organizes data into a meaningful order.

Common sorting approaches:

- Alphabetical (A–Z or Z–A)
- Numerical (Highest to Lowest or Lowest to Highest)
- Chronological (Oldest to Newest or vice versa)

Custom Sorting:

- Prioritizing a specific product or offering
- Grouping competitors together for comparison
- Highlighting key items at the top of a report

Custom sorting helps align the data presentation with business priorities.

15.3 Understanding Filtering

Filtering allows analysts to **drill down into the data** by showing only values that meet specific criteria.

What filtering enables:

- Viewing revenue for a single client
- Analyzing performance for a specific time period
- Focusing on a single geography or product line

Filtering removes distractions by hiding unnecessary rows without deleting any data.

15.4 Filtering and Visualization

Filtering is especially powerful when combined with visualizations.

Example:

- A bar chart showing monthly sales
- Apply a filter to view:
  - Sales for one country only, or
  - Sales for a specific product line

This allows the same chart to answer multiple questions without redesigning it.

15.5 Key Insights from Data Professionals

- Filtering and sorting help analysts "get to the heart of the data"
- They enable fast, focused decision-making
- Users can explore data without scrolling through large datasets
- Well-filtered views improve clarity and trust in insights

15.6 Summary

Filtering and sorting are not just technical tools; they are **analytical thinking tools**. They transform raw data into targeted insights and empower both analysts and decision-makers to find exactly what they need, when they need it.

16 Common Excel Functions for Data Analysis

After learning how to filter and sort data, the next essential step in analysis is using **functions** to derive insights from data. In this section, we focus on four of the most commonly used Excel functions: **IF, IFS, COUNTIF, and SUMIF**.

These functions help analysts classify data, summarize results, and support decision-making without writing long or error-prone formulas.

16.1 The IF Function

The **IF function** is a logical function that evaluates a condition and returns one value if the condition is **TRUE**, and another value if it is **FALSE**.

General Syntax:

```
=IF(logical_test, value_if_true, value_if_false)
```

Example 1: Order Shipped Status

Suppose we want to check whether an order has been shipped.

- Column G contains shipment status (e.g., "shipped", "pending")
- We add a new column named **Shipped?**

Formula in cell H2:

```
=IF(G2="shipped","Yes","No")
```

Explanation (line by line):

- Check if cell G2 contains the text "shipped"
- If TRUE, return "Yes"
- If FALSE, return "No"

Copying this formula down the column labels each order clearly.

16.2 Using IF for Value Classification

Example 2: Sales Size Classification

Suppose we want to identify whether a sales order is above or below \$3,000.

Formula in cell F2:

```
=IF(E2>3000,"Over 3k","Under 3k")
```

Explanation:

- Compare the sales value in E2 with 3000
- Assign a descriptive label instead of raw numbers

This makes reports easier to interpret for non-technical stakeholders.

16.3 Nested IF Functions

Sometimes more than two outcomes are required. This can be done by **nesting IF functions**.

Example: Order Size Categories

Logic:

- Sales $> 5000 \rightarrow$ Large
- Sales between 3000 and 5000 $\rightarrow$ Medium
- Sales $< 3000 \rightarrow$ Small

Nested IF Formula:

```
=IF(E2>5000,"Large",IF(E2>3000,"Medium","Small"))
```

Important Note: While Excel supports many nested IFs, they:

- Are difficult to read
- Are hard to maintain
- Increase error risk

16.4 The IFS Function (Recommended)

The **IFS function** simplifies multiple conditions into a cleaner formula.

General Syntax:

```
=IFS(condition1,value1, condition2,value2, ...)
```

Equivalent IFS Formula:

```
=IFS(E2>5000,"Large", E2>3000,"Medium", E2<=3000,"Small")
```

Advantages of IFS:

- Easier to read
- Easier to edit later
- Fewer parentheses

16.5 Combining IF with Conditional Formatting

Functions become more powerful when combined with **visual cues**.

Example: Retention Percentage

Step 1: Calculate retention

```
=Year_Resale_Value / Retail_Price
```

Format the result as a percentage.

Step 2: Label performance

Formula:

```
=IF(G2>0.69,"Good","Poor")
```

Step 3: Visual Highlighting

Using Conditional Formatting:

- "Good" → Green text, light green fill
- "Poor" → Red text, light red fill

This allows insights to be seen instantly.

16.6 The COUNTIF Function

COUNTIF counts how many cells meet a specific condition.

Syntax:

```
=COUNTIF(range, criterion)
```

Example: Orders by Country

Formula:

```
=COUNTIF(Country_Column,"UK")
```

Key Points:

- Text criteria must be in quotation marks
- Function is not case-sensitive

COUNTIFS

COUNTIFS allows multiple conditions (e.g., Country = UK AND Sales ¿ 3000).

16.7 The SUMIF Function

SUMIF adds values that meet a given condition.

Syntax:

```
=SUMIF(range, criterion, sum_range)
```

Example 1: Sum Large Orders

```
=SUMIF(Sales,">3000")
```

Explanation:

- Only sales greater than $3000 are added
- “¿3000” must be in quotes because it uses an operator

Example 2: Using Wildcards

```
=SUMIF(ProductLine,"*cars",Sales)
```

This sums sales for all product lines ending with “cars”.

SUMIFS

SUMIFS supports multiple criteria, such as:

- Country = USA
- Product Line = Classic Cars
- Sales ¿ 3000

16.8 Summary

- **IF / IFS** help classify and label data
- **COUNTIF** helps measure frequency
- **SUMIF** helps summarize financial values
- Combining functions with formatting improves insight delivery

These functions form the foundation of spreadsheet-based data analysis and are essential skills for any data analyst.

17 Lookup Functions in Excel: VLOOKUP and HLOOKUP

After learning logical and conditional functions such as IF, IFS, COUNTIF, and SUMIF, the next important category of Excel functions is **lookup (reference) functions**. Lookup functions allow us to retrieve related data from a separate table using a shared key.

In this section, we focus on **VLOOKUP** and **HLOOKUP**.

17.1 VLOOKUP Function

VLOOKUP stands for **Vertical Lookup**. It is used to search for a value in the **first column** of a table and return a corresponding value from another column in the same row.

Typical Use Case:

- Product price lookup
- Employee department lookup
- Customer information lookup

Syntax

```
=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
```

Parameter Explanation

- **lookup_value**: The value you are searching for
- **table_array**: The lookup table containing the data
- **col_index_num**: Column number (starting from 1) of the return value
- **range_lookup** (optional):
  - FALSE or 0 → Exact match
  - TRUE or 1 → Approximate match (default)

17.2 VLOOKUP Example: Car Price Lookup

Suppose we want to find the retail price of a car named *Corvette* from a car sales dataset.

Example Formula:

```
=VLOOKUP("Corvette", A2:G156, 5, FALSE)
```

Explanation (line by line):

- Search for "Corvette" in the first column of the table
- Look within the range A2:G156
- Return the value from the 5th column (Price column)
- Use FALSE to ensure an exact match

Important Rule: The lookup value column **must always be the leftmost column** of the lookup table.

17.3 Relative vs Absolute Cell References in VLOOKUP

When copying VLOOKUP formulas, incorrect results may occur due to **relative references**.

Problem

Copying the formula changes the lookup table reference.

Solution

Convert lookup table references to **absolute references** using dollar signs.

```
=VLOOKUP(V5, $A$2:$G$156, 5, FALSE)
```

Key Insight:

- Keep lookup table fixed (absolute reference)
- Allow lookup value to change (relative reference)

This ensures the formula works correctly when dragged or copied.

17.4 Data Connectivity Advantage

Once VLOOKUP is applied:

- Changing data in the main table
- Automatically updates values in the lookup results

This creates a **dynamic connection** between tables.

17.5 HLOOKUP Function

HLOOKUP stands for **Horizontal Lookup**. It works similarly to VLOOKUP but searches across the **top row** of a table instead of a column.

When to Use HLOOKUP

- When comparison values are arranged horizontally
- When lookup keys are stored in the first row

Syntax

```
=HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])
```

Difference from VLOOKUP:

- Uses `row_index_num` instead of column index
- Searches by row rather than column

17.6 HLOOKUP Example: Horsepower Rating

Suppose a lookup table contains horsepower levels in the top row:

- Low HP
- Medium HP
- High HP

Each column below contains a symbol rating.
Example Formula:

```
=HLOOKUP(K2, Y21:AA22, 2, FALSE)
```

Explanation:

- Look for the HP level in the top row
- Search within the lookup table
- Return the rating from row 2
- Use exact matching

Formatting the result with the **Wingdings font** converts symbols into visual ratings.

17.7 VLOOKUP vs HLOOKUP

- VLOOKUP is more commonly used
- Most datasets store lookup values vertically
- HLOOKUP is useful for compact or summary tables

17.8 Modern Alternative: XLOOKUP

XLOOKUP is a newer and improved function that:

- Replaces both VLOOKUP and HLOOKUP
- Works vertically and horizontally
- Does not require column or row index numbers

Note: XLOOKUP is available only in newer Excel versions (Excel 365, Web, and Mobile).

17.9 Summary

- **VLOOKUP** retrieves data by column
- **HLOOKUP** retrieves data by row
- Absolute references are critical for correct copying
- Lookup functions create dynamic data relationships

These functions are fundamental for real-world spreadsheet analysis and are widely used in business, finance, and data analytics.

18 Pivot Tables in Excel

After learning how to use `VLOOKUP` and `HLOOKUP`, we now move to one of the most powerful tools for data analysis in Excel: **Pivot Tables**. Pivot Tables allow analysts to summarize, analyze, and explore large datasets efficiently and from multiple perspectives.

18.1 Why Pivot Tables Are Important

While filters, formulas, and functions help answer many questions, they are often not sufficient for deeper analysis. Pivot Tables provide a dynamic and flexible way to:

- Summarize large volumes of data quickly
- Identify trends and patterns
- Compare values across categories
- Create presentable insights for stakeholders

A key advantage of Pivot Tables is that they are **dynamic**: when the source data changes or expands, the Pivot Table updates accordingly.

18.2 Preparing Data for Pivot Tables

Before creating a Pivot Table, ensure the dataset is in a suitable format:

- Format the data as a table for best results
- Ensure there is only one header row
- Remove blank rows and blank columns
- Ensure numeric fields are formatted as numbers
- Ensure date fields are formatted as dates

Formatting data correctly ensures accurate analysis and avoids common Pivot Table errors.

18.3 Formatting Data as a Table

Formatting data as a table improves structure and usability:

1. Select any cell within the dataset
2. Go to **Home** → **Format as Table**
3. Choose a table style
4. Confirm that **My table has headers** is selected

Once formatted:

- Filter buttons are automatically added to headers
- New rows are automatically included in the table
- Formatting is applied consistently

18.4 Creating a Pivot Table

To create a Pivot Table:

1. Select any cell in the table
2. Go to **Insert** → **PivotTable**
3. Confirm the table name or range
4. Choose to place the Pivot Table in a new worksheet

Excel opens a new worksheet with:

- A Pivot Table layout area
- The **PivotTable Fields** pane

18.5 Using Pivot Table Fields

Pivot Tables are built by dragging fields into four areas:

- **Rows**: Categories displayed vertically
- **Columns**: Categories displayed horizontally

- **Values**: Numerical data to summarize
- **Filters**: High-level filtering of the entire Pivot Table

For example:

- Manufacturer → Rows
- Model → Rows (below Manufacturer)
- Price → Values
- Unit Sales → Values

Fields can be rearranged or removed easily by dragging them between sections.

18.6 Formatting Pivot Table Values

By default, Pivot Table values are formatted as *General*. To improve readability:

1. Open **Value Field Settings**
2. Choose **Number Format**
3. Apply formats such as Currency or Percentage

This ensures numerical results are presented clearly and professionally.

18.7 Performing Calculations in Pivot Tables

Pivot Tables support calculated fields for advanced analysis:

- Go to **PivotTable Analyze** → **Fields, Items & Sets**
- Select **Calculated Field**
- Create formulas using existing fields

Example: To calculate total model sales:

$$\text{Total Sales} = \text{Price} \times \text{Unit Sales}$$

The calculated field appears as a new column in the Pivot Table and updates dynamically with the data.

18.8 Key Benefits for Data Analysts

Pivot Tables allow Data Analysts to:

- Analyze large datasets without writing complex formulas
- Generate insights quickly
- Create reports suitable for business presentations
- Adapt analysis instantly by changing fields

18.9 Summary

In this section, we learned:

- How to format data as a table
- How to create Pivot Tables
- How to organize data using Pivot Table fields
- How to perform calculations within Pivot Tables

In the next section, we will explore additional Pivot Table features to further enhance data analysis.